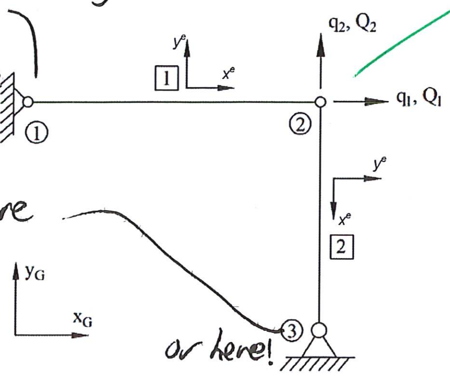
Example 1. Analysis of truss frames in global coordinate system

(1) A Truss structure pinned at supports is modelled using two bar elements:

Nodal point fully constrained and cannot translate due to pin so no DOF assigned here



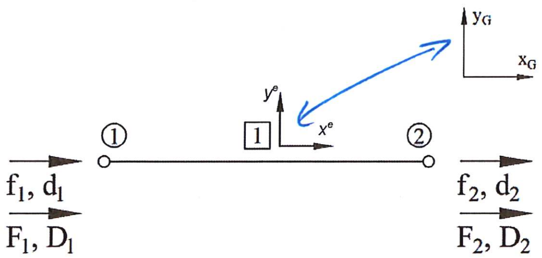
or here!

Potential for non-zero displacements in X^G and Y^G so 2 DOF here.

- The overall structure's motion/deformation can be described by **2 DOFs** (q_1 and q_2)
- Q_1 and Q_2 are the external loads applied at corresponding global DOFs (q_1 and q_2)
- Element #2 local coordinate axis (x^e, y^e) does **not** align with global coordinate axes (X_G, Y_G).

(2) Truss structures can be broken down into separate bar elements with corresponding local DOFs and nodal forces:

Element #1

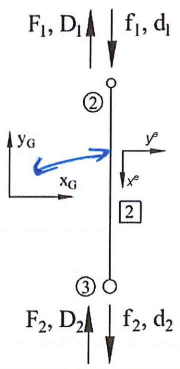


d_1 and d_2 – element nodal deformations in local coordinate axis system.
D_1 and D_2 – element nodal deformations in global coordinate axis system.
f_1 and f_2 – element nodal forces in local co-ordinates (axial forces for bar elements).
F_1 and F_2 – element nodal forces in global coordinate system.

Element #1 local (x^e, y^e) and global (X_G, Y_G) coordinate axes **match**, so:

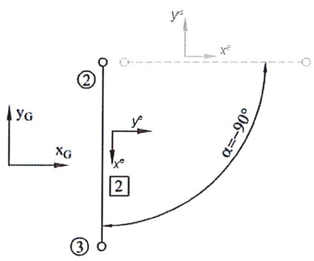
$$d_1 = D_1 \ \& \ f_1 = F_1$$
$$d_2 = D_2 \ \& \ f_2 = F_2$$

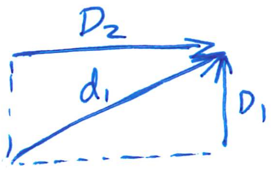
Element #2



- Element #2 local (x^e, y^e) and global (x_G, y_G) coordinate axes **do not match.**
- What do we do??
- Nothing matches and this is devolving into chaos!

(3) Note that Element #2 is merely rotated through -90deg.





D_1 and D_2 must add together in a vector sense to be equal to d_1

D_1 is X_G at node ①
D_2 is Y_G at node ①
D_3 is X_G at node ②
D_4 is Y_G at node ②.

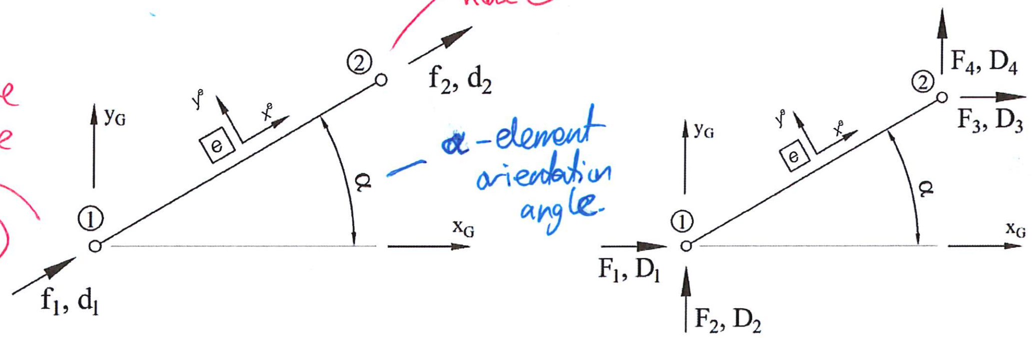
(4) For a general case we can write:

d_2 and f_2 are always aligned with x^e at node ②

Local coordinate system
lowercase f, d
<u>always</u> align with the element

Global coordinate system
uppercase F, D
<u>always</u> align with global coordinates (X_G, Y_G)

d_1 and f_1 are always in the direction of x^e at node ①



α - element orientation angle.

(5) We need to find a transformation matrix, which rotates the bar element by angle, α. Let:

$$\boxed{c = \cos\alpha \quad \& \quad s = \sin\alpha}$$

Hence we can find the relation between local element displacements d^e and nodal element displacement, D^e:

$$d_1 = D_1\cos(\alpha) + D_2\sin(\alpha) = c\,D_1 + s\,D_2$$
$$d_2 = D_3\cos(\alpha) + D_4\sin(\alpha) = c\,D_3 + s\,D_4$$

Or in matrix form:

$$\begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \overbrace{\begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 & 0 \\ 0 & 0 & \cos(\alpha) & \sin(\alpha) \end{bmatrix}}^{\text{Transformation matrix, which transforms between local (element) and global coordinates.}} \begin{pmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{pmatrix} = \Lambda^e\, D^e$$

Uppercase Lambda.

Where the matrix that 'transforms' the co-ordinate system is (unsurprisingly) called the transformation matrix:

$$\Lambda^e = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 & 0 \\ 0 & 0 & \cos(\alpha) & \sin(\alpha) \end{bmatrix}$$

(6) Element nodal displacements, D^e, and element nodal forces, F^e, (in global co-ordinates) can be transformed into corresponding displacements, d^e, and internal forces, f^e in local co-ordinates:

$$d^e{}_{\{2x1\}} = \Lambda^e{}_{\{2x4\}}\; D^e{}_{\{4x1\}}$$

$$F^e{}_{\{4x1\}} = \Lambda^{e^T}{}_{\{4x2\}}\; f^e{}_{\{2x1\}}$$

$$\Lambda^e{}_{\{2x4\}} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 & 0 \\ 0 & 0 & \cos(\alpha) & \sin(\alpha) \end{bmatrix}$$

Bar element transformation matrix, Λ^e, is general and can be applied to any bar element.

(7) What about the stiffness matrix K^e?

- If an element is rotated α degrees the original 2×2 stiffness matrix, K^e, is no longer valid as it must now be able to process pieces of the 4 (not 2) displacements, D^e, and forces, F^e.
- Element nodal forces in the global coordinate system can be found using the relation:

$$F^e = \widehat{K}^e D^e$$

Where D^e is a 4×1 vector representing element nodal displacements, and $\widehat{K}^e$ is 4×4 element global stiffness matrix.

- How can we transform element local stiffness matrix, $K^e_{\{2\times2\}}$, into element global stiffness matrix, $\widehat{K}^e_{\{4\times4\}}$?

(8) Recall,

$$K^e_{\{2\times2\}} = \int_0^L \Psi'(x)^T EA\, \Psi'(x) dx$$

where Ψ represents element shape functions in local coordinates (x^e, y^e) for $u(x) = \Psi(x)\, d$.

(9) Hence, in global coordinates (x_G, y_G) element displacement at any point x can be calculated:

$$\hat{u} = \Psi(x)\, d^e = \Psi(x)\, (\Lambda\, D^e) = (\Psi(x)\, \Lambda)\, D^e$$

(10) Therefore, in formulating element stiffness matrix in global coordinates we can write:

$$\widehat{K}^e_{\{4\times4\}} = \int_0^L \Lambda^T \Psi'(x)^T EA\, \Psi'(x) \Lambda\, dx = \Lambda^T \Bigg(\underbrace{\int_0^L \Psi'(x)^T EA\, \Psi'(x)\, dx}_{K^e_{2\times2}} \Bigg) \Lambda$$

element stiffness matrix in local coordinates.

element stiffness matrix in global coordinates

$$\boxed{\begin{gathered} \widehat{K}^e_{\{4\times4\}} = \Lambda^{e^T}_{\{4\times2\}} K^e_{\{2\times2\}} \Lambda^e_{\{2\times4\}} \\ K^e_{\{2\times2\}} = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \\ \Lambda_{\{2x4\}} = \begin{bmatrix} c & s & 0 & 0 \\ 0 & 0 & c & s \end{bmatrix} \\ \text{recalling the shorthand} \\ c = \cos\alpha \quad \& \quad s = \sin\alpha \end{gathered}}$$

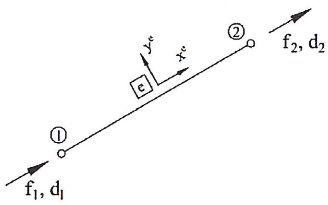
In the Computer Lab this week, enter the matrix using this information, not this.

Or

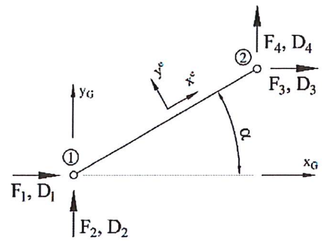
$$\boxed{\widehat{K}^e_{\{4\times4\}} = \Lambda^{e^T}_{\{4\times2\}} K^e_{\{2\times2\}} \Lambda^e_{\{2\times4\}} = \frac{EA}{L}\begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix}}$$

Converting between element deflections in local and global coordinates

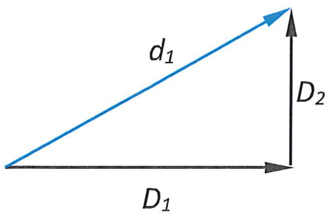
Element DOF in local (element) co-ordinates

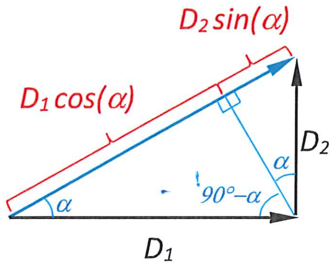


Element DOF in global co-ordinates



Let's look at the vector sum of displacements, D_1^e and D_2^e, which must be equal to d_1^e





$d_1 = D_1\cos(\alpha) + D_2\sin(\alpha)$

And similarly:

$d_2 = D_3\cos(\alpha) + D_4\sin(\alpha)$

And in matrix form, this is expressed as $d^e = \Lambda^e D^e$, which can be written as:

$$\begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \overbrace{\begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 & 0 \\ 0 & 0 & \cos(\alpha) & \sin(\alpha) \end{bmatrix}}^{\Lambda^e} \begin{pmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{pmatrix}$$

However, if you instead want to transform from local co-ordinates d^e to global co-ordinates, D^e, the equation is:

$D^e = {\Lambda^e}^T d^e$ - But why is it this equation? Why do we use ${\Lambda^e}^T$, why not the inverse, ${\Lambda^e}^{-1}$?

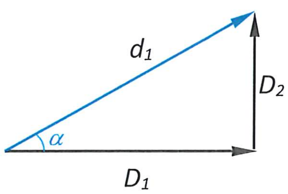
Well, first up, Λ^e is rectangular, not square, so the inverse doesn't exist. That's a good reason to not use the inverse, but why the transpose? Does that actually work? Well, let's take a look...

$$\begin{pmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{pmatrix} = \overbrace{\begin{bmatrix} \cos(\alpha) & 0 \\ \sin(\alpha) & 0 \\ 0 & \cos(\alpha) \\ 0 & \sin(\alpha) \end{bmatrix}}^{\Lambda^{eT}} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix}$$

If we write out these equations, we get:

$$D_1 = d_1\cos(\alpha) + 0d_2$$
$$D_2 = d_1\sin(\alpha) + 0d_2$$
$$D_3 = 0d_1 + d_2\cos(\alpha)$$
$$D_4 = 0d_1 + d_2\sin(\alpha)$$

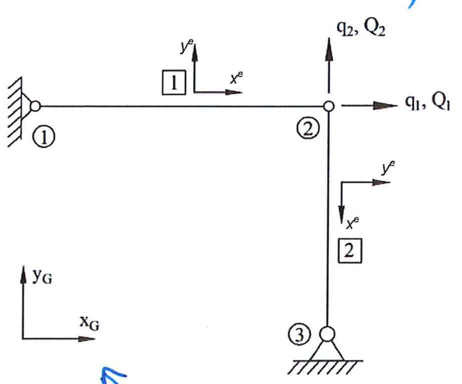
Recall our displacement vector triangle (which can also be applied to node 2).



Example 1 Revisited. Assembly + Analysis of Plane (2D) Trusses

DOF Assignment refer back to p33

- Global structure
- Identify global (structure-level) DOFS, q
- Identifying external loads, Q

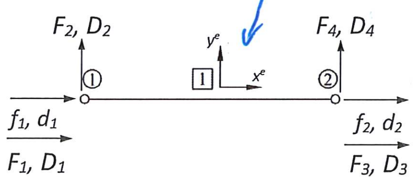


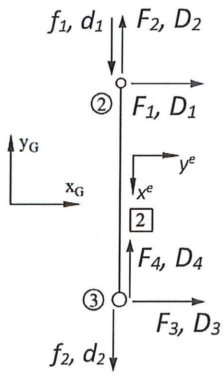
The coordinate axes align $\alpha^1 = 0°$.

Element Freebody Diagrams: **Element #1** **Element #2**

Break structure into elements

Identify element DOFs, d
- Local axes, (x^e, y^e)
- Angle of rotation, α





Transformation matrix

$$\Lambda = \begin{bmatrix} c & s & 0 & 0 \\ 0 & 0 & c & s \end{bmatrix}$$

$$c = \cos\alpha$$

$$s = \sin\alpha$$

$$\alpha = 0° \qquad c = \cos\alpha = \cos(0°) = 1 \qquad s = \sin\alpha = \sin(0°) = 0$$

$$\Lambda^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\alpha = -90° \text{ or } +270° \qquad c = \cos(-90°) = 0 \qquad s = \sin(-90°) = -1$$

$$\Lambda^2 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Superscript refers to the element number.

Element local deformation

$$d^1 = \begin{pmatrix} d_1^1 \\ d_2^1 \end{pmatrix} = \Lambda^1 D^1 = \begin{pmatrix} D_1^1 \\ D_3^1 \end{pmatrix} \qquad d^2 = \begin{pmatrix} d_1^2 \\ d_2^2 \end{pmatrix} = \Lambda^2 D^2 = \begin{pmatrix} -D_2^2 \\ -D_4^2 \end{pmatrix}$$

subscripts refer to the DOF # within that element.

d_2^1 is the second local degree-of-freedom in local coordinates for element $\boxed{1}$

Element nodal forces

$$F^1 = \begin{pmatrix} F_1^1 \\ F_2^1 \\ F_3^1 \\ F_4^1 \end{pmatrix} = (\Lambda^1)^T f^1 = \begin{pmatrix} f_1^1 \\ 0 \\ f_2^1 \\ 0 \end{pmatrix} \qquad F^2 = \begin{pmatrix} F_1^2 \\ F_2^2 \\ F_3^2 \\ F_4^2 \end{pmatrix} = (\Lambda^2)^T f^2 = \begin{pmatrix} 0 \\ -f_1^2 \\ 0 \\ -f_2^2 \end{pmatrix}$$

We now have information that explains how the degrees of freedom for each element, in local co-ordinates (d_1,d_2) correspond to the degrees of freedom for each element in global co-ordinates (D_1, D_2, D_3, and D_4)

We can use the information obtained on the previous page to formulate the element stiffness equation in global co-ordinates for each element, using the follow equations:

$K^e_{\{2\times2\}} = \frac{EA}{L}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$	$\widehat{K}^e_{\{4\times4\}} = {\Lambda^e}^T_{\{4\times2\}} K^e_{\{2\times2\}} \Lambda^e_{\{2\times4\}}$	$F^e = \widehat{K}^e\, D^e$

However, that will only get us so far, as the equation for each element is still independent from other elements, does not include information about how these elements connect to each other, and does not include information about support points/boundary conditions.

We need to solve the overall structure as a system, but how do we do that?

How do we get from element-level equations using F^e, D^e & $\widehat{K}^e$ to structure-level equations that use Q, q, and K_G?

Solving global structure for motions	$Q_{\{n\times1\}} = K_{G\{n\times n\}}\, q_{\{n\times1\}}$ K_G- $n\times n$ global structure stiffness matrix, where n is the number of global DOFs in structure

The answer is using **assembly matrices**, A^e, for each element.

$$D^e_{\{4\times1\}} = (A^e)^T_{\{4\times n\}}\, q_{\{n\times1\}}$$

$$Q^e_{\{n\times1\}} = \sum_{e=1}^{n_e} A^e{}_{\{n\times4\}}\, F^e_{\{4\times1\}}$$

$$K_{G\{n\times n\}} = \sum_{e=1}^{n_e} A^e{}_{\{n\times4\}}\ \widehat{K}^e_{\{4\times4\}}\ (A^e)^T{}_{\{4\times n\}}$$

n_e – total number of elements

n – total number of global DOF (the q's)

That looks easy (maybe)! But, what is this assembly A^e and how does it map local element displacements and forces in global coordinates (D^e, F^e) into final model DOFs (the q's)??

- Since it "maps" force and displacement it is made of 0s and 1s only.

⇒ Assembly matrices contain only 1s and zeros, so they don't scale anything, they just assign to appropriate locations.

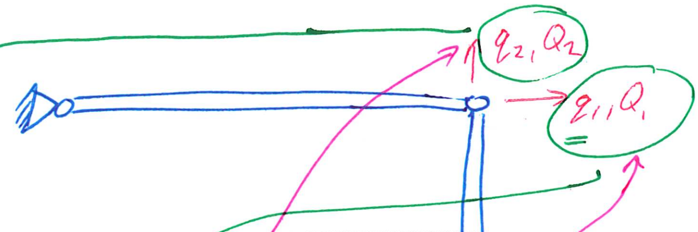
In this example, in the global level the structure has 2 DOFs, where each element has 4 DOFs. Hence, the assembly matrices in this example will be 2×4 matrix containing 0s and 1s.

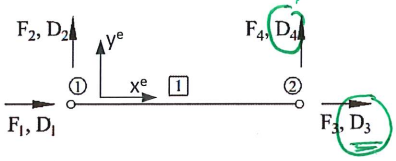
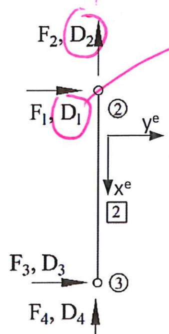
columns = # element DOFs in global co-ordinates — for a given element type. Always equal to 4 for a bar element.

rows = # global DOFs in the overall structure (the # of q's)

Element Assembly Matrix

In this specific problem it's two. But this can vary with different structures.



	Element #1	Element #2
Element DOF, D^e, and nodal forces, F^e		

Formulation of assembly matrices

$$A^1 = \begin{array}{c} \\ q_1 \\ q_2 \end{array} \begin{array}{cccc} D_1^1 & D_2^1 & D_3^1 & D_4^1 \\ \end{array}$$
$$A^1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

which means, $D_3^1 = q_1$ and $D_4^1 = q_2$

$$A^2 = \begin{array}{c} \\ q_1 \\ q_2 \end{array} \begin{array}{cccc} D_1^2 & D_2^2 & D_3^2 & D_4^2 \\ \end{array}$$
$$A^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

which means, $D_1^2 = q_1$ and $D_2^2 = q_2$

D_3^2 and D_4^2 are fully constrained.

(6) Applied external force is equal to the sum of element forces:

$$Q = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = \sum_{e=1}^{n_e} A^e F^e = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} F_1^1 \\ F_2^1 \\ F_3^1 \\ F_4^1 \end{pmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{pmatrix} F_1^2 \\ F_2^2 \\ F_3^2 \\ F_4^2 \end{pmatrix}$$

$$Q = \begin{pmatrix} F_3^1 + F_1^2 \\ F_4^1 + F_2^2 \end{pmatrix}$$

(7) The overall structure's stiffness matrix (referred to as the Global Stiffness Matrix K_G) can be found by assembling element global stiffness matrices:

K_G - the overall global stiffness matrix that represents the full structure.
All information about all elements and their connectivity into the structure.

$$K_G = \sum_{e=1}^{n_e} A^e \hat{K}^e (A^e)^T = \underbrace{A^1 \hat{K}^1 (A^1)^T}_{K_G^1} + \underbrace{A^2 \hat{K}^2 (A^2)^T}_{K_G^2}$$

Note the difference between A^e assembly matrix and Λ^e which is the transformation matrix.

(8) And recall that element global stiffness matrix is defined:

$$\hat{K}^e_{\{4\times4\}} = \Lambda^{eT}_{\{4\times2\}} K^e_{\{2\times2\}} \Lambda^e_{\{2\times4\}}$$

$$\hat{K}^e_{\{4\times4\}} = \frac{EA}{L} \begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix}$$

$c = \cos\alpha$ and $s = \sin\alpha$

Element #1	Element #2
$\alpha = 0^o$	$\alpha = -90^o$ or $+270^o$.
$\widehat{K}^1 = \frac{EA}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\widehat{K}^2 = \frac{EA}{L}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}$

(9) Element global stiffness matrix contribution to the total stiffness matrix of the whole structure is:

Element #1	Element #2
$K_G^1 = A^1\widehat{K}^1{A^1}^T$	$K_G^2 = A^2\widehat{K}^2{A^2}^T$
$K_G^1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\frac{EA}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$	$K_G^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}\frac{EA}{L}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$
$K_G^1 = \frac{EA}{L}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$K_G^2 = \frac{EA}{L}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

(10) Total, overall, global stiffness matrix (KG) that represents the stiffness of the overall structure, including all elements and the boundary conditions (support points). This is simply obtained by summing up the contribution from each element.

Overall K_G is always square with dimension $n \times n$ where n = # of global degrees of freedom of the structure. (the q values)

Assuming $E^1 = E^2 = E$, $A^1 = A^2 = A$, $L^1 = L^2 = L$.

$$K_G = K_G^1 + K_G^2 = \frac{EA}{L}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{EA}{L}\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \frac{EA}{L}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(11) Now structure can be solved for displacements:

$$\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = K_G \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \frac{EA}{L}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{pmatrix} q_1 \\ q_2 \end{pmatrix}$$

$$\begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = K_G^{-1}\, Q = \underbrace{\frac{L}{EA}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{K_G^{-1}}\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = \frac{L}{EA}\begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix}$$

This is the point at which the system is "solved". This is the step where we simultaneously calculate all DOF for the structure.

Any subsequent analysis is called "post-processing".

Which can be separated out to:

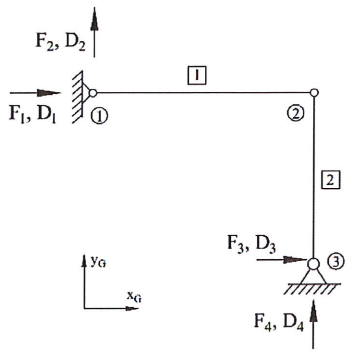
$$\boxed{q_1 = \frac{L}{EA} Q_1} \qquad \boxed{q_2 = \frac{L}{EA} Q_2}$$

This answer for q_1 relies only on Element 1 boundary conditions. Element 2 is perpendicular to Element 1 and provides no resistance to the q_1 DOF as it is a bar element having no flexural stiffness. Similarly, the displacement in the q_2 DOF is a function of Q_2 and stiffness of Element #2 only.

Essentially, the matrix equation reduces down to two independent equations, each of which is a simple axial force-deformation equation.

The equations are only independent and separable due to the 90° rotation between them and the fact that we are dealing with pin-jointed bar elements.

(12) What about reaction loads?



What are the reaction forces F_1, F_2 at joint #1?

for element $\boxed{1}$.

What are the reaction forces F_3, F_4 at joint #3?

for element $\boxed{2}$.

(13) We know:

$$F^e = \widehat{K}^e D^e$$

where $\widehat{K}^e$ is element stiffness matrix in global co-ordinates and D_e is the vector of element nodal displacements, again in global co-ordinates.

To use the equation above, we need to know what deflections have occurred within each element. Thankfully, the assembly matrix that we have already defined can be used to 'extract' or 'select' the relevant pieces of the overall (now solved) deflection vector, q.

$$D^e = (A^e)^T q$$

solved global deflections.

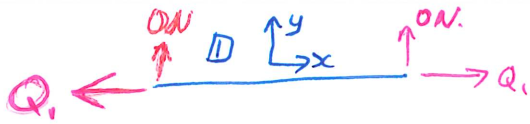
Assembly matrix both "builds-up" our system, but also "breaks it down"!

hence, the element nodal forces can be found:

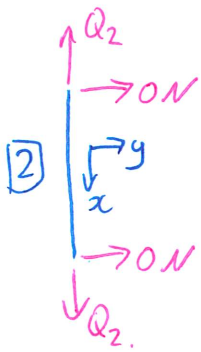
$$F^e = \widehat{K}^e (A^e)^T q$$

Nodal forces for element #1:

$$F^1 = \begin{pmatrix} F_1^1 \\ F_2^1 \\ F_3^1 \\ F_4^1 \end{pmatrix} = \widehat{K}^1 (A^1)^T q = \frac{EA}{L} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{Q_1 L}{EA} \\ \frac{Q_2 L}{EA} \end{bmatrix} = \begin{pmatrix} -Q_1 \\ 0 \\ Q_1 \\ 0 \end{pmatrix}$$



Nodal forces for element #2:



$$F^2 = \begin{pmatrix} F_1^2 \\ F_2^2 \\ F_3^2 \\ F_4^2 \end{pmatrix} = \widehat{K}^2 (A^2)^T q = \frac{EA}{L} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{Q_1 L}{EA} \\ \frac{Q_2 L}{EA} \end{bmatrix} = \begin{pmatrix} 0 \\ Q_2 \\ 0 \\ -Q_2 \end{pmatrix}$$

Some Final General Comments

Assembly matrix, A^e, maps model global DOF (q_i) to element DOF in global coordinates (D_i). As this step also defines the total number of structural DOFs, the assembly matrix applies the boundary conditions (support points) as well.

In the previous example we get 2 global DOF (q_1, q_2) and global element displacements in global coordinates, $D^e = (D_1^e \quad D_2^e \quad D_3^e \quad D_4^e)^T$

$A^e \quad =$

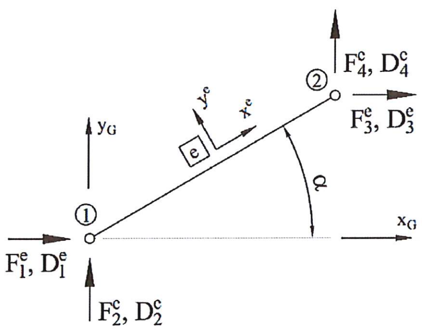
	D_1^e	D_2^e	D_3^e	D_4^e
q_1				
q_2				

Where each row i of the assembly matrix A^e is constructed:

$$\left.\begin{array}{l} 0 = q_i \neq D_i \\ 1 = q_i = D_i \end{array}\right\}$$

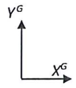
Therefore, only 0 or 1 in each field. Each row will be all zeros or have one "1" as each possible movement/DOF, q_i, cannot equal more than one D_i.

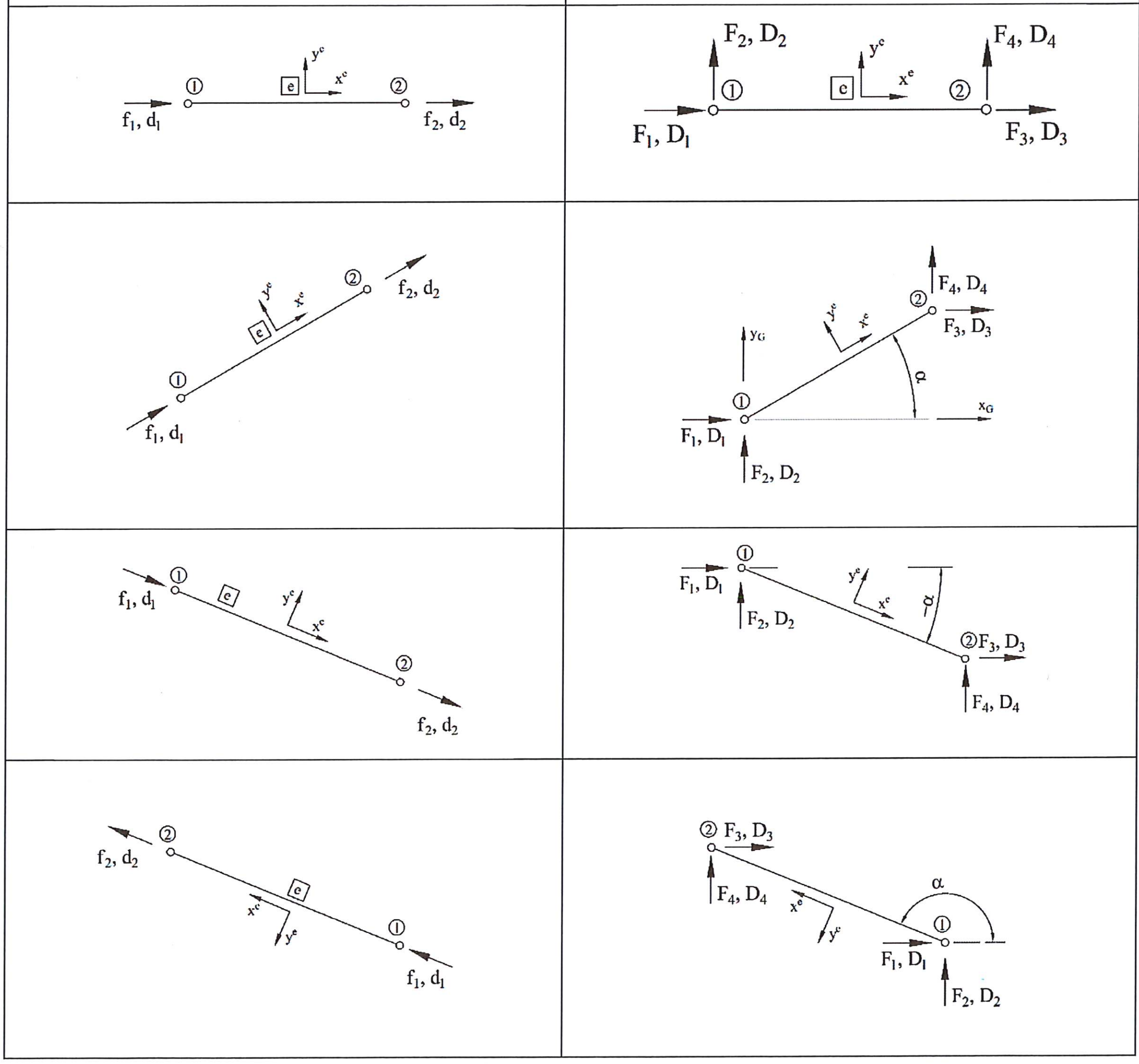
When labelling D_i, F_i recall we must match global coordinates. Hence, for generic case shown below, $X_G \rightarrow D_1^e, D_3^e$ and $Y_G \rightarrow D_2^e, D_4^e$. **This is a fixed nomenclature and approach.**



Local vs Global Element Degrees of Freedom for a Bar Element

Element DOF in local (element) co-ordinates	Element DOF in global co-ordinates
Local x^e at node 1 → d_1 Local x^e at node 2 → d_2	Global X^G at node 1 → D_1 Global Y^G at node 1 → D_2 Global X^G at node 2 → D_3 Global Y^G at node 2 → D_4





$\alpha^e = 0°$

$\alpha^e = 35°$

$\alpha^e = +335°$
or $\alpha^e = -25°$

$\alpha^e = 155°$

Note: you can choose any element end as a start node #1, which will define f_1, d_1. The angle of rotation, α, will depend on the choice of start node #1 and end node #2.

# of q's relates to the number of nodal points and the number of support points.

Problem 1 summary: All major steps and equations.

(1) Define possible motions/degrees-of-freedom of the overall structure: $q = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix}$

and external forces applied to the overall structure: $Q = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix}$

(2) Define element displacement-deformation relationship by creating local and global stiffness matrices for each element:

Local coordinate system	**Global coordinate system**
$f^e{}_{\{2\times1\}} = K^e{}_{\{2\times2\}}\, d^e{}_{\{2\times1\}}$	$F^e{}_{\{4\times1\}} = \widehat{K}^e{}_{\{4\times4\}}\, D^e{}_{\{4\times1\}}$
$K^e{}_{\{2\times2\}} = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$	$\widehat{K}^e{}_{\{4\times4\}} = \Lambda^{e^T}{}_{\{4\times2\}}\, K^e_{\{2\times2\}}\, \Lambda^e{}_{\{2\times4\}}$
	$\Lambda^e{}_{\{2\times4\}} = \begin{bmatrix} c & s & 0 & 0 \\ 0 & 0 & c & s \end{bmatrix}$

(3) Define element assembly matrix, A^e, (by hand) for each element:

Number of element DOF in global co-ordinates

alway 4 for a bar element.

$A^e =$

	D_1^e	D_2^e	D_3^e	D_4^e
q_1				
q_2				
…				
q_n				

Number of structure DOF

(4) Assemble the total global stiffness matrix of the whole system:

$$K_{G\,\{n\times n\}} = \sum_{e=1}^{n_e} A^e{}_{\{n\times4\}}\, \widehat{K}^e_{\{4\times4\}}\, (A^e)^T{}_{\{4\times n\}}$$

n_e – total number of elements

n – total number of DOF of the overall structure

(5) Solve the system for motions/displacements:

$$Q = K_G\, q$$

$$\downarrow$$

$$q = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = (K_G)^{-1}\, Q$$

or np.linalg.solve.

(6) Find the reaction forces:

$$F^e = \widehat{K}^e\, D^e = \widehat{K}^e (A^e)^T q$$

where

$$D^e = (A^e)^T q$$

(7) Find element motions:

$$d^e = \Lambda^e\, D^e$$

and element nodal forces:

$$F^e = (\Lambda^e)^T\, f^e$$

Summary of Different Stiffness Matrices

There are several levels of stiffness matrix, which can potentially be confusing. This table is included to help you understand the significance of the different forms of stiffness matrix and how additional information about the structure is introduced as we go through the process to build up the overall global stiffness matrix, K_G, that defines the overall structure, including information about stiffness contributions from every element and the boundary conditions.

Different Stiffness Equations	Different forms of Stiffness Matrices	Information included/not included: Information on Element Material and Geometric Properties	Information on Element Orientation	Information on how the element is connected to other elements and/or to support points	Information on more than one element
$f^e = K^e d^e$ K^e relates element forces (in element coordinates), f^e to resulting deflections (in element coordiantes) d^e for one element	K^e Element stiffness matrix in local (element) co-ordinates	YES ✓ E, A, L.	NO ×	NO ×	NO ×
$F^e = \hat{K}^e D^e$ $\hat{K}^e$ relates element forces (in global coordinates) F^e to resulting deflections (in global coordiantes) D^e for one element	$\hat{K}^e = \Lambda^{e^T} K^e \Lambda^e$ Element stiffness matrix in global co-ordinates	YES ✓	YES ✓ Introduced via the transformation matrix	NO ×	NO ×
No stiffness equation using just K_G^e This is an intermediate step only. We do not solve any equations using just K_G^e, we need to generate the overall K_G first	$K_G^e = A^e \ \hat{K}^e \ (A^e)^T$ This element's contribution to the global stiffness matrix	YES ✓	YES ✓	YES ✓ Introduced through the assembly matrix	NO ×
$Q = K_G \, q$ K_G relates overall structure-level applied external forces (in global coordinates), Q to resulting overall structure-level deflections (in global coordiantes), q	$K_G = \sum_{e=1}^{n_e} K_G^e$ The overall global stiffness matrix. Captures all the stiffness terms of the total structure	YES ✓	YES ✓	YES ✓	YES ✓ Information on every element and every support point.

Let's consider a larger pin-jointed truss made up of ten bar elements that is supported by a two pin-joints at the left edge of the structure. We will assume that every element is made of the same homogenous material and that all elements have the same cross-sectional area, expressed mathematically as:

$$E^1 = E^2 = E^3 = E^4 = E^5 = E^6 = E^7 = E^8 = E^9 = E^{10} = E$$

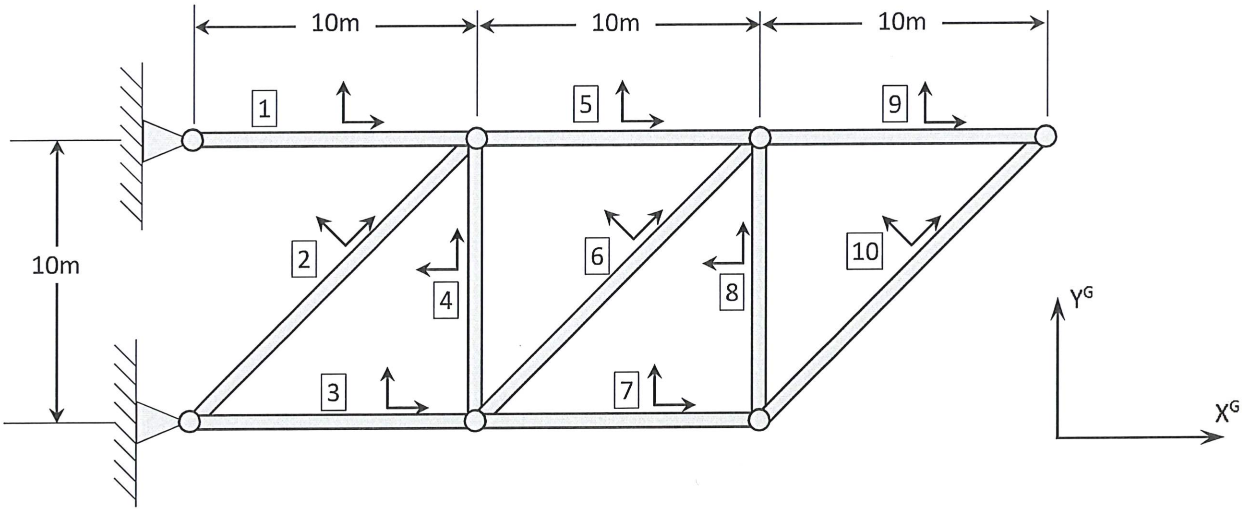
$$A^1 = A^2 = A^3 = A^4 = A^5 = A^6 = A^7 = A^8 = A^9 = A^{10} = A$$

We will assume that the truss includes a square grid, so that the lengths of many elements are the same:

$$L^1 = L^3 = L^4 = L^5 = L^7 = L^8 = L^9 \approx 10m$$

and

$$L^2 = L^6 = L^{10} = 14.1m$$



As the local stiffness matrix in local coordinates (K^e) depends only on only geometric (A and L) and material (E) properties (and not element orientation or connectivity within the structure), then:

$$K^1 = K^3 = K^4 = K^5 = K^7 = K^8 = K^9$$

and

$$K^2 = K^6 = K^{10}$$

However, we will now introduce element orientation information (via the transformation matrix) when we calculate the element stiffness matrix in global coordinate ($\widehat{K}^e = \Lambda^{e^T} K^e \Lambda^e$). Elements with the same geometric and material properties **and** the same orientation will have the same $\widehat{K}$

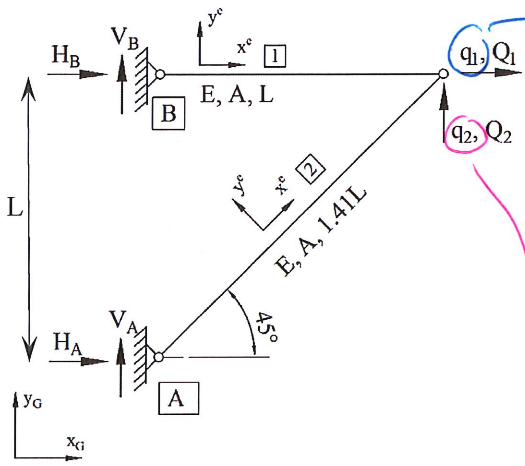
$\widehat{K}^1 = \widehat{K}^3 = \widehat{K}^5 = \widehat{K}^7 = \widehat{K}^9$	$\widehat{K}^4 = \widehat{K}^8$	$\widehat{K}^2 = \widehat{K}^6 = \widehat{K}^{10}$

Once we introduce the element connectivity information through the assembly matrices (and noting that the assembly matrix for every element will be unique as every element connects into the structure differently), then the contribution of each element to the global stiffness matrix ($K_G^e = A^e\ \widehat{K}^e\ (A^e)^T$) will be unique, such that:

$$K_G^1 \neq K_G^2 \neq K_G^3 \neq K_G^4 \neq K_G^5 \neq K_G^6 \neq K_G^7 \neq K_G^8 \neq K_G^9 \neq K_G^{10}$$

Problem 2

External forces $Q_1 = 0\ kN$, $Q_2 = 100kN$ act on the structure below. Find the structural deflections, the reaction forces at supports A & B, and element internal forces. Assume $L_1 = 10m$ and $L_2 = 14.1m$, and both members are made of 100mm diameter solid steel bars with elastic modulus, $E = 200$ GPa.



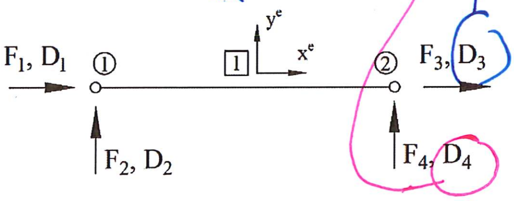
Solution

Finding local element stiffness matrix for each element:

Element #1 Free-body Diagram:

$q_3 = D'_3$

$\alpha' = 0°$



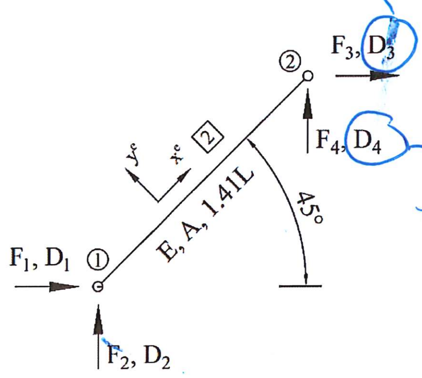
$q_2 = D'_4$

$$K^1 = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\widehat{K}^1 = \frac{EA}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Element #2

$$\boxed{q_1 = D_3^2}$$



$$\boxed{q_2 = D_4^2}$$

$$K^2 = \frac{EA}{1.41L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\widehat{K}^2 = \frac{EA}{1.41L}\begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix}$$

(3) Creating assembly matrix for each element:

$A^1 =$

	D_1^1	D_2^1	D_3^1	D_4^1
q_1	0	0	1	0
q_2	0	0	0	1

$A^2 =$

	D_1^2	D_2^2	D_3^2	D_4^2
q_1	0	0	1	0
q_2	0	0	0	1

(4) Assembling global element matrices to form a total stiffness matrix:

$$K_G = \underbrace{A^1\,\widehat{K}^1\,(A^1)^T}_{K_G^1} + \underbrace{A^2\,\widehat{K}^2\,(A^2)^T}_{K_G^2}$$

$$K_G = \frac{EA}{L}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{EA}{1.41L}\begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} = \frac{EA}{L}\begin{bmatrix} 1.35 & 0.35 \\ 0.35 & 0.35 \end{bmatrix}$$

(5) Solving the structure for deflections:

Given in the question

$$Q = K_G\, q$$

$$Q = \begin{pmatrix} Q_1 \\ Q_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 100{,}000 \end{pmatrix} = \frac{EA}{L}\begin{bmatrix} 1.35 & 0.35 \\ 0.35 & 0.35 \end{bmatrix}\begin{pmatrix} q_1 \\ q_2 \end{pmatrix}$$

$$K_G^{-1} = \frac{L}{EA}\begin{bmatrix} 1 & -1 \\ -1 & 3.86 \end{bmatrix}$$

Simply we np.linalg.solve.

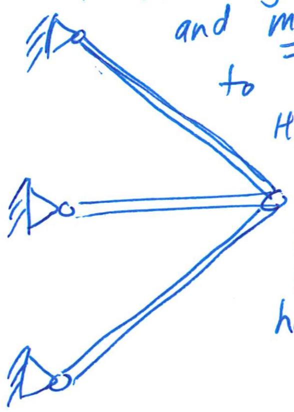
Hence, the deflection is equated as:

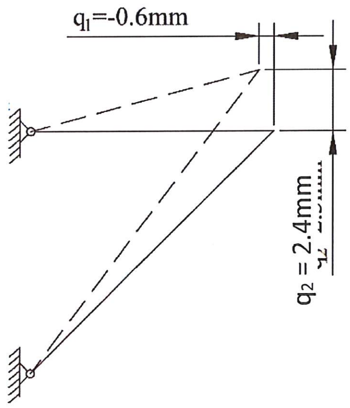
$$q = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = K_G^{-1}\, Q = \frac{L}{EA}\begin{bmatrix} 1 & -1 \\ -1 & 3.86 \end{bmatrix}\begin{pmatrix} 0 \\ 100{,}000 \end{pmatrix}$$

$$\begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{pmatrix} -0.0006m \\ 0.0024m \end{pmatrix} = \begin{pmatrix} -0.6366mm \\ 0.2432mm \end{pmatrix}$$

These are the same values obtained manually (with a lot of effort) on pages 18-25

The presence of this third element makes the system statically indeterminate and much harder to solve by hand. However, through FEA this problem is not much harder at all.





no annoying quadralaterals

no need to manually interpret geometry.

Element #1 is **compressed** as $q_1 = -0.6366mm < 0$

Element #2 is **stretched** as $q_2 = 2.432mm > 0$

$D^e = A^{eT} @ q$

(6) Finding element nodal forces: $F^e = \widehat{K}^e\, D^e = \widehat{K}^e (A^e)^T q$

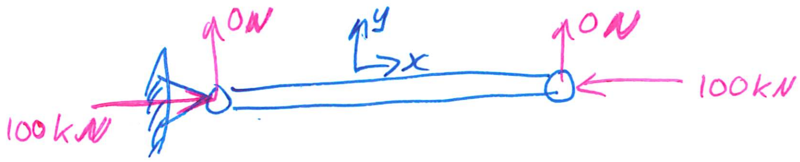
The assembly matrix extracts the relevant pieces of the solved q vector and assigns them into the appropriate place with D^e.

<u>Element #1</u>

$$F^1 = \begin{pmatrix} F_1^1 \\ F_2^1 \\ F_3^1 \\ F_4^1 \end{pmatrix} = \widehat{K}^1 (A^1)^T q = \frac{EA}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -0.006 \\ 0.0024 \end{pmatrix}$$

$$\begin{pmatrix} F_1^1 \\ F_2^1 \\ F_3^1 \\ F_4^1 \end{pmatrix} = \begin{pmatrix} 1e^5 \\ 0 \\ -1e^5 \\ 0 \end{pmatrix}$$

Element $\boxed{1}$ FBD:

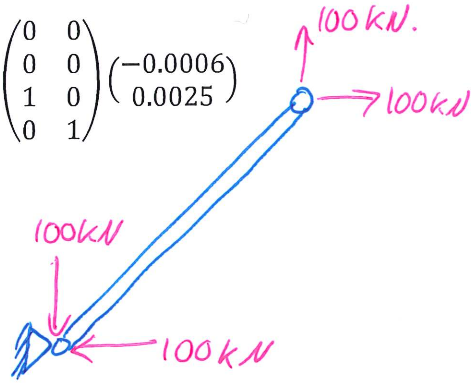


Where reaction forces at support B, H_B & V_B, are represented by element #1 nodal forces F_1^1 & F_2^1, respectively:

$$H_B = F_1^1 = 100kN$$
$$V_B = F_2^1 = 0$$

<u>Element #2:</u>

$$F^2 = \begin{pmatrix} F_1^2 \\ F_2^2 \\ F_3^2 \\ F_4^2 \end{pmatrix} = \widehat{K}^2 (A^2)^T q = \frac{EA}{1.41L} \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \\ -0.5 & -0.5 & 0.5 & 0.5 \end{bmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -0.0006 \\ 0.0025 \end{pmatrix}$$



$$\begin{pmatrix} F_1^2 \\ F_2^2 \\ F_3^2 \\ F_4^2 \end{pmatrix} = \begin{pmatrix} -1e^5 \\ -1e^5 \\ 1e^5 \\ 1e^5 \end{pmatrix}$$

Where reaction forces at support A, H_A & V_A, are represented by element #2 nodal forces F_1^2 & F_2^2:

$$H_A = F_1^2 = -100kN$$
$$V_A = F_2^2 = -100kN$$

Quick sanity check.
This was pinned so d_1' must be zero.

(7) Finding local element deformation, d:

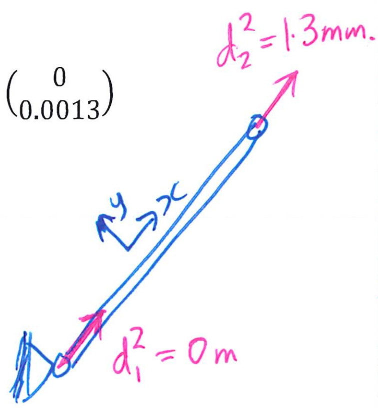
$$d^e = \Lambda^e \cdot D^e = \Lambda^e \cdot \underbrace{(\Lambda^e)^T q}_{D^e}$$

<u>Element #1 deformation:</u>

$$d^1 = \Lambda^1 \, (\Lambda^1)^T q = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -0.0006 \\ 0.0025 \end{pmatrix} = \begin{pmatrix} 0 \\ -0.0006 \end{pmatrix} \begin{matrix} m \\ m \end{matrix}$$

Element #2 deformation:

$$d^2 = \Lambda^2 \, (\Lambda^2)^T q = \begin{bmatrix} 0.707 & 0.707 & 0 & 0 \\ 0 & 0 & 0.707 & 0.707 \end{bmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -0.0006 \\ 0.0025 \end{pmatrix} = \begin{pmatrix} 0 \\ 0.0013 \end{pmatrix}$$



Hence, element #1 **compressed** by:

$$\Delta L^1 = d_2^1 - d_1^1 = -0.0006 - 0 = -0.0006m \rightarrow \boldsymbol{-0.6mm}$$

And element #2 **stretched** by:

$$\Delta L^2 = d_2^2 - d_1^2 = 0.0013 - 0 = 0.0013m \rightarrow \boldsymbol{1.3mm}$$

(8) Finding element strains: $\varepsilon^e = \frac{\Delta L^e}{L^e}$

$d_2^e - d_1^e = \Delta L.$

Element #1 strain:

$$\varepsilon^1 = \frac{\Delta L^1}{L^1} = \frac{-0.0006}{10} = -6e^{-5} = -60\mu\varepsilon$$

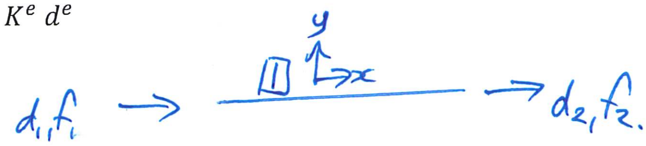
Element #2 strain:

$$\varepsilon^2 = \frac{\Delta L^2}{L^2} = \frac{0.0013}{14.1} = 9e^{-5} = +90\mu\varepsilon$$

(9) Finding element forces in local (element) co-ordinates:

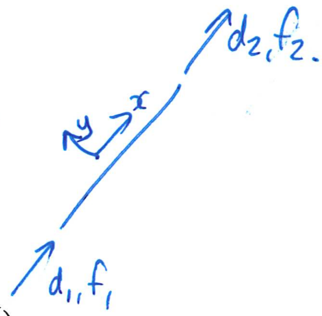
$$f^e = K^e\, d^e$$

Element #1 local forces:



$$f^1 = K^1\, d^1 = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{pmatrix} 0 \\ -0.0006 \end{pmatrix} = \begin{pmatrix} 1e^5 \\ -1e^5 \end{pmatrix} N = \begin{pmatrix} 100\ kN \\ -100\ kN \end{pmatrix}$$

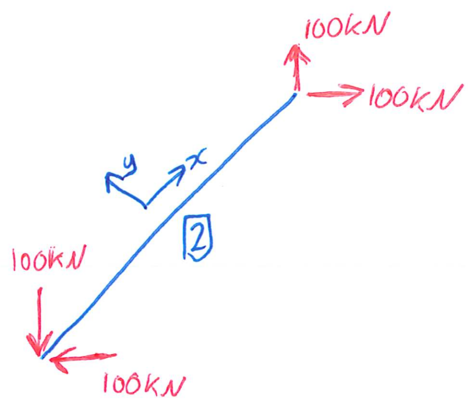
Element #2 local forces:

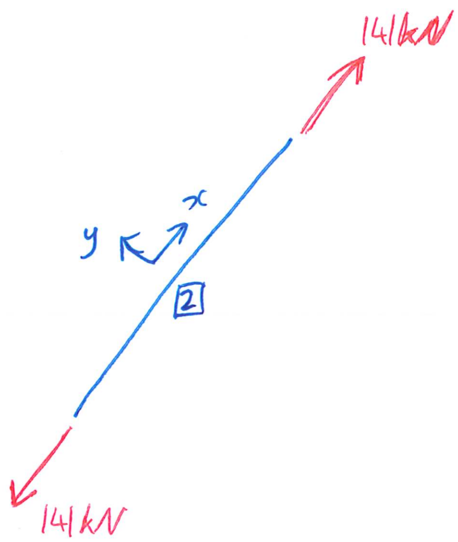


$$f^2 = K^2\, d^2 = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{pmatrix} 0 \\ 0.0013 \end{pmatrix} = \begin{pmatrix} -1.41e^5 \\ 1.41e^5 \end{pmatrix} N = \begin{pmatrix} -141\ kN \\ 141\ kN \end{pmatrix}$$

We have the following force information for Element #2:

$$F^2 = \begin{pmatrix} F_1^2 \\ F_2^2 \\ F_3^2 \\ F_4^2 \end{pmatrix} = \begin{pmatrix} -100kN \\ -100kN \\ +100kN \\ +100kN \end{pmatrix} \qquad f^2 = \begin{pmatrix} -141\ kN \\ 141\ kN \end{pmatrix}$$





How to code it?

(1) Create functions for bar elements to calculate the bar element stiffness matrix in local co-ordinates, K^e, and the bar element stiffness matrix in global co-ordinates, $\widehat{K}^e$.

(2) Create assembly matrix, A^e, for each element by hand.

(3) Assemble total stiffness matrix of the whole system by:

$$K_{G\{n\times n\}} = \sum_{e=1}^{n_e} A^e{}_{\{n\times 4\}} \widehat{K}^e_{\{4\times 4\}} \, (A^e)^T{}_{\{4\times n\}}$$

$$n_e - \text{total number of elements}$$

$$n - \text{total number of overall (structural) DOF}$$

(4) Solve equation $Q = K_G \cdot q$ to get displacements, q. Use the matrix inverse or np.linalg.solve

(5) Calculate element nodal forces in global co-ordinates

$$F^e = \widehat{K}^e \, D^e = \widehat{K}^e (A^e)^T q$$

for any given element so you can pick out and solve reaction loads.

(6) Calculate element nodal displacements in local co-ordinates:

$$d^e = \Lambda^e \, D^e = \Lambda^e \, (A^e)^T \, q$$

(7) Calculate element internal forces in local co-ordinates:

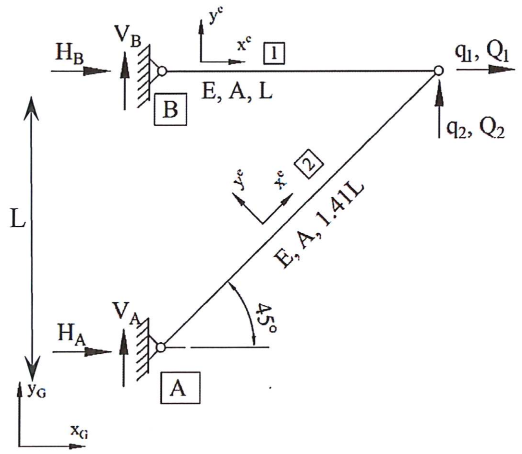
$$f^e = K^e \, d^e$$

and strains:

$$\varepsilon^e = \frac{d_2^e - d_1^e}{L^e}$$

Problem 2 solved using Python

Given the bars are 100mm diameter solid steel with $E = 200$ GPa, and $L = 10m$, and external forces $Q_1 = 0$, $Q_2 = 100kN$, find the reaction forces at supports A & B, and element internal forces.



The functions *local_bar* and *global_bar* are **not** inbuilt Python commands – you need to write these. Given geometric and material properties, E, A, and L, **for one element** the *local_bar* code needs to generate the element stiffness matrix in local coordinates **for that one element** (refer to page 35). You can then call this function as many times as you have elements within your structure.

The function *global_bar* is then written to take the output from *local_bar* and the element orientation angle, α, generate the transformation matrix, Lambda (Λ) and then generate the element stiffness matrix in global coordinates, $\widehat{K}$. Again, refer to page 35 for supporting equations.

The angle α in Python is expected as radians into cos and sin functions. np.radians( ).

Take K^e and angle α and getting transformation matrix and $\hat{K}e$

Element 1	Element 2
$K1 = local\_bar(E, A, L) =$ $K^1 = 1 \times 10^8 \times \begin{bmatrix} 1.5708 & -1.5708 \\ -1.5708 & 1.5708 \end{bmatrix}$	$K2 = local\_bar(E, A, L) =$ $K^2 = 1 \times 10^8 \times \begin{bmatrix} 1.1140 & -1.1140 \\ -1.1140 & 1.1140 \end{bmatrix}$
$K1hat, Lambda1 = global\_bar(K1, \alpha 1)$ $\widehat{K}^1 = 1 \times 10^8$ $\begin{vmatrix} 1.5708 & 0 & -1.5708 & 0 \\ 0 & 0 & 0 & 0 \\ -1.5708 & 0 & 1.5708 & 0 \\ 0 & 0 & 0 & 0 \end{vmatrix}$ $Lambda1 = \Lambda^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	$K2hat, Lambda2 = global\_bar(K2, \alpha 2)$ $\widehat{K}^2 = 1 \times 10^7$ $\begin{vmatrix} 5.5702 & 5.5702 & -5.5702 & -5.5702 \\ 5.5702 & 5.5702 & -5.5702 & -5.5702 \\ -5.5702 & -5.5702 & 5.5702 & 5.5702 \\ -5.5702 & -5.5702 & 5.5702 & 5.5702 \end{vmatrix}$ $\Lambda^2 = \begin{bmatrix} 0.7071 & 0.7071 & 0 & 0 \\ 0 & 0 & 0.7071 & 0.7071 \end{bmatrix}$
Assembly Matrix (generated manually): $A^1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Assembly Matrix (generated manually): $A^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
$K_G^1 = A^1 \widehat{K}^1 (A^1)^T =$ $K_G^1 = 1 \times 10^8$ $\begin{vmatrix} 1.5708 & 0 \\ 0 & 0 \end{vmatrix}$	$K_G^2 = A^2 \widehat{K}^2 (A^2)^T =$ $K_G^2 = 1 \times 10^7$ $\begin{vmatrix} 5.5702 & 5.5702 \\ 5.5702 & 5.5702 \end{vmatrix}$

Finding total stiffness matrix:

$$K_G = K_G^1 + K_G^2$$

$$K_G = 1 \times 10^8 \times \begin{vmatrix} 2.1278 & 0.5570 \\ 0.5570 & 0.5570 \end{vmatrix}$$

Creating external force vector (values given in the question):

$$Q = \begin{Bmatrix} 0\,N \\ 100{,}000\,N \end{Bmatrix}$$

Solving equation for motions:

$$Q = K_G\, q \rightarrow q = inv(K_G)\, Q \ or \ \underline{np.linalg.solve}$$

$$q = \begin{bmatrix} -0.0006\,m \\ 0.0024\,m \end{bmatrix}$$

Element #1 nodal force vector (in global co-ordinates):

$$F^1 = \widehat{K}^1 D^1 = \widehat{K}^1\,((A^1)^T q) = \begin{pmatrix} 100{,}000 \\ 0 \\ -100{,}000 \\ 0 \end{pmatrix}$$

Element #2 nodal force vector (in global co-ordinates):

$$F^2 = \widehat{K}^2 D^2 = \widehat{K}^2\,((A^2)^T q) = \begin{pmatrix} -100{,}000 \\ -100{,}000 \\ 100{,}000 \\ 100{,}000 \end{pmatrix}$$

Refer back to the FBD on pages 49/50. to interpret.

Element #1 nodal displacement vector (in local co-ordinates):

$$d^1 = \Lambda^1\,(A^1)^T q = \begin{pmatrix} 0 \\ -0.6366 \end{pmatrix} \times 10^{-3}$$

where $\Lambda^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ is by definition given, using α = 0°, and used in the second step when generating K1hat

Element #2 nodal displacement vector (in local co-ordinates):

$$d^2 = \Lambda^2\,(A^2)^T q = \begin{pmatrix} 0 \\ 0.0013 \end{pmatrix}$$

$\Lambda^2 = \begin{bmatrix} 0.7071 & 0.7071 & 0 & 0 \\ 0 & 0 & 0.7071 & 0.7071 \end{bmatrix}$ is by definition given, using α = 45°, and used in the second step when generating K2hat

Element #1 strain:

$$\varepsilon^1 = \frac{d_2^1 - d_1^1}{L^1} = \frac{(-0.6366 - 0) \times 10^{-3}}{10} = -6.36 \times 10^{-5}$$

Element #2 strain:

$$\varepsilon^2 = \frac{d_2^2 - d_1^2}{L^2} = \frac{(0.0013 - 0)}{14.1} = 9 \times 10^{-5}$$

Element #1 nodal force vector (in local co-ordinates):

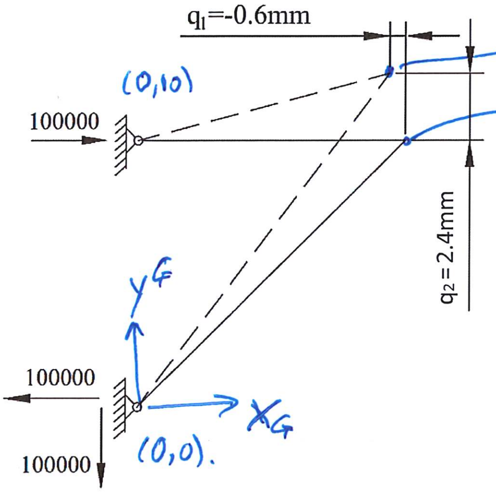
$$f^1 = K^1 d^1 = \begin{pmatrix} 100{,}000 \\ -100{,}000 \end{pmatrix}$$

Element #2 nodal force vector (in local co-ordinates):

$$f^2 = K^2 d^2 = \begin{pmatrix} -1.4142 \\ 1.4142 \end{pmatrix} \times 10^5$$

define some magnification factor.
$\Rightarrow$ disp_mag = 100
exaggerated deflection position $(10 + disp\_mag * q_1,\ 10 + disp\_mag * q_2)$.

Plotting a deformed shape and reaction forces:



deflected position $(10 + q_1,\ 10 + q_2)$
undeflected = (10, 10).

In Python: axes.plot([x_1, x_2],[y_1, y_2])

```
axes.plot([0, 10],[10, 10],
          label = 'Undeflected
                   element 1').
```

```
axes.plot([0, 10+q1],[10, 10+q2],
          label= "Deflected
                  Element 1")
```

What if I wanted to know the internal deflection within an element based upon these nodal deflections?
What about a position halfway along element $\boxed{1}$?

$$u(x=0.5L') = u(x=5m) = \left(1-\frac{5}{10}\right)(0) + \left(\frac{5}{10}\right)(-0.6366\ \text{mm})$$

$$= -0.3183\ \text{mm}$$

(axial deflection internally half way along element $\boxed{1}$).

$$u(x) = \psi_1(x)\,d_1 + \psi_2(x)\,d_2$$

$$u(x) = \left(1-\frac{x}{L}\right)d_1 + \frac{x}{L}\,d_2$$

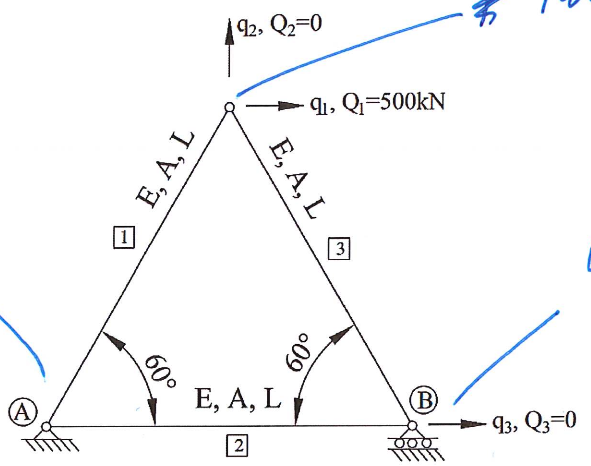
Problem 3

A truss made up of three bar elements is supported by a pin at point A and a roller at point B. A horizontal load of 500kN is applied at the top of the structure. Assume all members are $L = 5m$ long, have a circular hollow section with 100mm outside diameter and 10mm wall thickness, and are made of aluminium with $E = 70GPa$. Solve for structural deflections, element forces and deflections, and determine the support reaction forces.

(1) Define possible motions/DOF, q, and identify external forces, Q:

Two potential non-zero deflections, two q values.

Pin constrains motion in X_G and Y_G direction.
⇒ no allowable deflection
⇒ no q values assigned here.

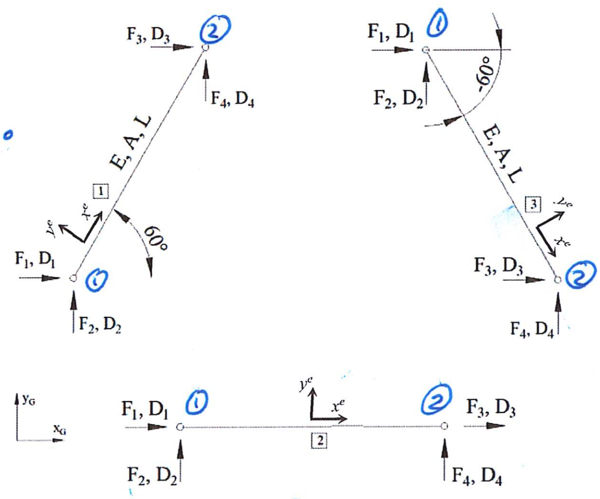


Vertical (Y_G) direction is constrained, so no q value in that direction.
⇒ However, permissible deflection in X_G
So one q value there.

(2) Defining global element DOF, D, and nodal forces, F on element free-body diagrams.

$\alpha^1 = +60°$

$$\boxed{D_3^1 = q_1 \quad D_4^1 = q_2}$$



$\alpha^3 = +300°$
$= -60°$

$$\boxed{D_1^3 = q_1 \quad D_2^3 = q_2 \quad D_3^3 = q_3}$$

$\alpha^2 = 0°$

$$\boxed{D_3^2 = q_3}$$

(3) Defining global element stiffness matrix, $\hat{K}^e$, for each element:

Element #1 ($\alpha^1 = 60°$):

$$\hat{K}^1 = \frac{EA}{L}\begin{bmatrix} 0.25 & 0.43 & -0.25 & -0.43 \\ 0.43 & 0.75 & -0.43 & -0.75 \\ -0.25 & -0.43 & 0.25 & 0.43 \\ -0.43 & -0.75 & 0.43 & 0.75 \end{bmatrix}$$

Element #2 ($\alpha^2 = 0°$):

$$\hat{K}^2 = \frac{EA}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Element #3 ($\alpha^3 = -60°$):

$$\hat{K}^3 = \frac{EA}{L}\begin{bmatrix} 0.25 & -0.43 & -0.25 & 0.43 \\ -0.43 & 0.75 & 0.43 & -0.75 \\ -0.25 & 0.43 & 0.25 & -0.43 \\ 0.43 & -0.75 & -0.43 & 0.75 \end{bmatrix}$$

(4) Define element assembly matrices:

Two all-zero columns. These DOF (D_1^1 and D_2^1) do not have a global/structural DOF which means that they're fixed.

Element #1

$A^1 =$

	D_1^1	D_2^1	D_3^1	D_4^1
q_1	0	0	1	0
q_2	0	0	0	1
q_3	0	0	0	0

There is no direct connection between element $\boxed{1}$ and q_3.

Element #2

$A^2 =$

	D_1^2	D_2^2	D_3^2	D_4^2
q_1	0	0	0	0
q_2	0	0	0	0
q_3	0	0	1	0

element $\boxed{2}$ doesn't directly connect to the node with q_1 and q_2.

Element #3

$A^3 =$

	D_1^3	D_2^3	D_3^3	D_4^3
q_1	1	0	0	0
q_2	0	1	0	0
q_3	0	0	1	0

all zero column due to the constraint from the roller which preved Y_G movement.

(5) Obtaining total stiffness matrix, K_G, by assembling global stiffness matrix of each element, $\widehat{K}^e$:

$$K_G = \sum_{e=1}^{3} A^e \ \widehat{K}^e \ (A^e)^T$$

$$K_G = A^1\widehat{K}^1(A^1)^T \ + \ A^2\widehat{K}^2(A^2)^T \ + \ A^3\widehat{K}^3(A^3)^T$$

Element #1 contribution to the total stiffness matrix:

$$K_G^1 = A^1\widehat{K}^1(A^1)^T = \frac{EA}{L}\begin{bmatrix} 0.25 & 0.43 & 0 \\ 0.43 & 0.75 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Element #2 contribution:

$$K_G^2 = A^2\widehat{K}^2(A^2)^T = \frac{EA}{L}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Element #3 contribution:

$$K_G^3 = A^3\widehat{K}^3(A^3)^T = \frac{EA}{L}\begin{bmatrix} 0.25 & -0.43 & -0.25 \\ -0.43 & 0.75 & 0.43 \\ -0.25 & 0.43 & 0.25 \end{bmatrix}$$

Hence, the total stiffness matrix is equal to a sum of element contributions:

$$K_G = K_G^1 + K_G^2 + K_G^3 = \frac{EA}{L}\begin{bmatrix} 0.5 & 0 & -0.25 \\ 0 & 1.5 & 0.43 \\ -0.25 & 0.43 & 1.25 \end{bmatrix}$$

(6) Solving the systems for deformations:

$$Q = K_G q \quad \rightarrow \quad q = K_G^{-1} Q$$

np.linalg.solve.
What is Q?
Given in the question:
$Q = \begin{Bmatrix} 500,000\ N \\ 0\ N \\ 0\ N \end{Bmatrix}$.

$$q = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \frac{L}{EA}\begin{pmatrix} 1125 \\ -72 \\ 250 \end{pmatrix} \times 10^3 \ m$$

(7) Finding element nodal forces:

$$F^e = \widehat{K}^e \ D^e = \widehat{K}^e (A^e)^T q$$

Element #1 nodal forces (in global co-ordinates):

$$F^1 = \widehat{K}^1(A^1)^T q = \begin{pmatrix} -250,000 \\ -433,000 \\ 250,000 \\ 433,000 \end{pmatrix} N$$

Element #2 nodal forces (in global co-ordinates):

$$F^2 = \widehat{K}^2(A^2)^T q = \begin{pmatrix} -250,000 \\ 0 \\ 250,000 \\ 0 \end{pmatrix} N$$

Element #3 nodal forces (in global co-ordinates):

$$F^3 = \widehat{K}^3(A^3)^T q = \begin{pmatrix} 250,000 \\ -433,000 \\ -250,000 \\ 433,000 \end{pmatrix} N$$

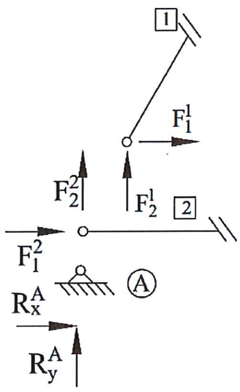
(8) Reaction forces at supports:

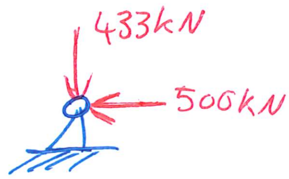
$F^1 = \begin{pmatrix} -250{,}000 \\ -433{,}000 \\ 250{,}000 \\ 433{,}000 \end{pmatrix} N$	$F^2 = \begin{pmatrix} -250{,}000 \\ 0 \\ 250{,}000 \\ 0 \end{pmatrix} N$	$F^3 = \begin{pmatrix} 250{,}000 \\ -433{,}000 \\ -250{,}000 \\ 433{,}000 \end{pmatrix} N$

Reaction forces can be found by superimposing corresponding global element forces, F^e.

Reaction forces at support A:

$$R_x^A = F_1^1 + F_1^2 = -250000 + (-250000) = -50000N = -500kN$$
$$R_y^A = F_2^1 + F_2^2 = -433000 + 0 = -433000N = -433kN$$





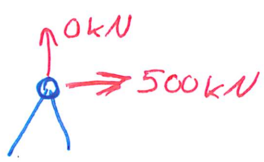
Reaction forces at support B:

$$R_x^B = F_3^2 + F_3^3 = 250000 + (-250000) = 0N$$
$$R_y^B = F_4^2 + F_4^3 = 0 + 433000 = 433000N = 433kN$$

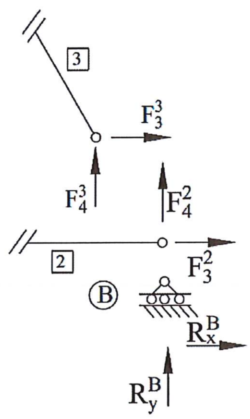
Look at the top node:

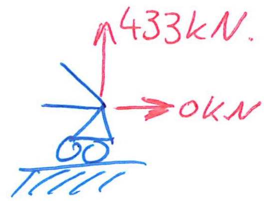
$$F^1_{3-4} = \begin{Bmatrix} 250\ 000 \\ 433\ 000 \end{Bmatrix} \begin{matrix} N \\ N \end{matrix}$$

$$F^2_{1-2} = \begin{Bmatrix} 250\,000\ N \\ -433\ 000\ N \end{Bmatrix}.$$



We see that the internal element forces at this node sum up to the applied external loads.





Quick sanity check:

The roller is incapable of providing a horizontal reaction, so $R_x^B = 0kN$ as expected.

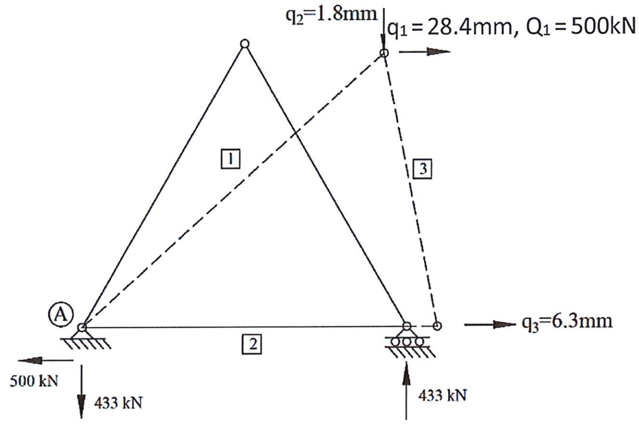
(9) Plotting deformed shape and reaction loads:

System deformations are calculated in Step (6). Recall that all members are $L = 5m$ long, circular hollow core 100mm outside diameter and 10mm wall thickness, made of aluminium, $E = 70GPa$, system deformation equals:

$$q = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} = \frac{L}{EA}\begin{pmatrix} 1125 \\ -72 \\ 250 \end{pmatrix} m = \frac{5}{(70 \times 10^9)(2.83 \times 10^{-3})}\begin{pmatrix} 1125 \\ -72 \\ 250 \end{pmatrix} = \begin{pmatrix} 0.0284 \\ -0.0018 \\ 0.0063 \end{pmatrix} m$$

where the cross sectional area of a hollow core tube is:

$$A = \frac{\pi}{4}(D_e^2 - D_i^2) = \frac{\pi}{4}(0.1^2 - (0.1 - 2 \times 0.01)^2) = 2.83 \times 10^{-3} m^2$$



(10) Calculating local element deformations: $d^e = \Lambda^e D^e = \Lambda^e (\Lambda^e)^T q$

Element #1 to #3 local deformations:

$$d^1 = \begin{pmatrix} 0 \\ 500{,}000 \end{pmatrix}\frac{L}{EA} = \begin{pmatrix} 0 \\ 0.0126 \end{pmatrix} m$$

$$d^2 = \begin{pmatrix} 0 \\ 250{,}000 \end{pmatrix}\frac{L}{EA} = \begin{pmatrix} 0 \\ 0.0063 \end{pmatrix} m$$

$$d^3 = \begin{pmatrix} 625{,}000 \\ 125{,}000 \end{pmatrix}\frac{L}{EA} = \begin{pmatrix} 0.0158 \\ 0.0032 \end{pmatrix} m$$

(11) Finding bar element strains and stresses:

$$\varepsilon^e = \frac{d_2^e - d_1^e}{L^e} \qquad\qquad \sigma^e = E^e \varepsilon^e$$

Element #1 to #3 strains and stresses: $\varepsilon^1 = \frac{d_2^1 - d_1^1}{L^1} = \frac{0.0126-0}{5} = 0.0025$

$$\sigma^1 = E^1\,\varepsilon^1 = (70 \times 10^9)(0.0025) = 176.8\ MPa$$

$$\varepsilon^2 = 0.0013 \rightarrow \sigma^2 = 88.4\ MPa$$

$$\varepsilon^3 = -0.0025 \rightarrow \sigma^3 = -176.8\ MPa$$